Intent-Based Service Provisioning and Closed-Loop Automation for Cobot Service Migration in a Multi-Stakeholder Environment

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***Abstract*—This paper presents a Proof-of-Concept (PoC) focused on intent-based service provisioning and closed-loop automation for collaborative robot (cobot) use case. The PoC demonstrates the integration of two key IBN enablers: the Intent-Based Network Intent Management Entity (IBN-IME) and CL Automation and Coordination, both aligned with 3GPP and ETSI Zero-Touch Service and Network Management (ZSM) standards. These enablers support the seamless interpretation of user intents, service deployment, and migration. The testbed, designed to emulate certain functionalities present in a 6G end-to-end system, highlights the potential for automation in future networks. The work addresses the need for an ecosystem that allows for the interaction of multiple stakeholders in a secure platform. The PoC presented can adapt to increasingly complex use cases, laying the groundwork for future improvements and broader deployments.**

***Index Terms*—Cobots, intent-based networks, zero-touch network and service management (ZSM), robot operating system (ROS), proof-of-concept (PoC).**

I. Introduction

IN RECENT discussions surrounding the sixth generation (6G) mobile network architecture, numerous proposals and innovative approaches have emerged, highlighting the growing expectations for communication networks to transcend traditional boundaries. The evolution of upcoming technologies, along with societal and environmental concerns, is reshaping the development of communication networks. There is a clear motivation to integrate diverse technologies and disciplines to meet the future demands of consumers. This requires a shift in focus from Key Performance Indicators (KPIs) to Key Value Indicators (KVIs), from a metrics-driven development (the 5G cornerstones, URLLC, mMTC, and eMBB) towards more holistic solutions that cater to emerging and future-driven use cases. In this work, we emphasize the "robots to cobots" family of use cases, which involve collaborative robots (cobots) and autonomous systems interacting with humans [1]. In general, cobot use cases extend across a wide range of applications such as domestic settings, or industrial environments [2], [3].

As an European flagship project, the Hexa-X-II consortium is contributing to shaping the 6G evolution by defining the design principles and developing a comprehensive 6G architecture. Building on the 6G vision in [4], this work aligns with the proposed system blueprint and addresses key ideas it presents, with particular emphasis on the following aspects:

- The native support for automation and distributed intelligence in 6G networks. That involves, the deployment of pervasive AI frameworks that enable predictive orchestration and optimization without human interaction.
- The integration of cloud platforms and computing services through the design of flexible interfaces between network and application modules. This requires support for APIs that expose 6G platform capabilities.
- The demonstration of 6G system capabilities by developing use cases such as collaborative robots (cobots).

Intent-Based Networks (IBN) are emerging as the future framework for managing and orchestrating 6G systems. The Hexa-X-II project, identifies core elements required for IBN implementation in 6G, highlighting the challenges posed by the coexistence of multiple stakeholders and domains [5]. This paper is closely aligned with the IBN framework presented in Hexa-X-II [6]. For the demonstrations carried out in the following sections, we utilize modules that implement the functionalities of two selected 6G technological enablers: Closed-Loop (CL) Platform implementing the CL Zero-touch Network and Service Management (ZSM) enabler and the IBN Intent Management Entity (IBN-IME), which includes functionalities for Intent Translation and Provisioning.

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Our main goal is to advance the progress towards 6G E2E systems by developing a testbed that implements Intent-Based service management modules and evaluating its performance. Developing a PoC can give significant insights into implementation challenges that cannot be foreseen in the architectural design process.

The main contributions of our work can be summarized as follows:

- Design and Development of 6G Capabilities in a Testbed: We develop a proof-of-concept (PoC) that integrates NLP-driven and intent-based service management for orchestrating a surveillance cobot use case. The testbed includes cloud computing nodes and applications orchestrated within an intent-based framework.
- AI-driven intent processing: We evaluate an AI-based approach for translating and decomposing intents. Specifically, we assess the role and capabilities of Natural Language Processing (NLP) interfaces in orchestrating 6G services.
- Closed-loop control for cobot operations: We define and validate closed-loop control mechanisms that enable fine-grained management of cobot tasks, supported by hierarchical state machines. This novel framework leverages AI and cloud orchestration to enhance cobot operation and management.

The remaining sections of this article are organized as follows. In Section II, the context of the research conducted and the motivation for this work is described. In Section III, the cobot use case is introduced, and main platform enablers, their architecture and algorithms are detailed. In Section IV, the experimental setup is introduced, along with the cobot infrastructure that support the experiments. In Section V, the results from the experiments carried out in the testbed are discussed. We outline the limitations and future enhancements needed for achieving a full working solution in Section VI. Finally, we summarize the key finding of our work in Section VII.

II. Background and Related Work

In [2], it is stated the demands from multiple sectors for a paradigm shift on the capabilities of mobile communication networks. The next generation is expected to consider technological advancements, social and economic demands, and environmental concerns.

Such a paradigm will raise the complexity of telecommunication systems, by incorporating diverse hardware and software resources and incorporating non-traditional communication services such as computing and sensing. In [7] deep architectural changes are foreseen. These changes must accommodate compute intensive tasks such as AI and allow to securely manage data from complex and diverse domains. The paper envisions the incorporation of an independent computing plane (in addition to the Data and Control Plane) to support dynamic service deployment across heterogeneous and large-scale systems. This reinforces the idea that a transition from Communication Service Providers (CSP) to Technology Companies (TechCos) is to come [8]. This is, network operators and CSPs are to collaborate to deliver end-to-end (E2E) services [9]. For instance, the concept of network slices established in 5G, which is passed onto 6G, is to expand into "digital slices" to accommodate the use cases that are driving the 6G development (e.g., collaborative robots, autonomous driving, digital healthcare, etc.). For such systems to succeed, the collaboration among multiple stakeholders must be consolidated. Systems will interface with each other through Application Programming Interfaces (API) and complex processes will have to be automated to enable an efficient management workflow [10]. Additionally, for the infrastructure's situational awareness assessment, monitoring, and diagnostic processes will need to be deployed across multiple layers of the management and orchestration stack.

In order to address these challenges, ZSM approaches developed by the European Telecommunications Standards Institute (ETSI) have been proposed [11]. The ZSM framework promotes collaboration between standardization bodies and incorporates data-driven AI/ML algorithms for closed-loop automation. Achieving this higher level of automation requires a significant transformation of traditional Network Management Systems (NMS).

IBN is one of the latest advancements in network operations, which is built upon the foundational concepts introduced by Software Defined Networking (SDN). IBN offers a single management plane capable of managing the entire network, regardless of its size or complexity. It is key to be aware that the management plan delivered by IBN aims to be protocol-agnostic and device-agnostic, IBN to function seamlessly in a multi-vendor and heterogeneous environment. To reach the objective of a more autonomous system and offer a more simplified network management (for the business users), a key aspect is the standardization of the functionalities that IBN systems should deliver. To this end, entities such as TMForum [12], 3GPP [13], [14] have presented their contributions that are being standardized by the ETSI [15]. The potential for Intent-Based frameworks and technologies has, therefore, been well identified and studied, yet their adoption is still slow.

The work presented herein focuses on how the combination of IBN and Closed Loops technologies bring benefits to achieving a more autonomous management of the services and the network resources. These two technologies are not the only possibilities for developing autonomous management systems, other technologies already used in the literature are possible. Table I presents a set of multiple autonomous systems management with their strengths and weaknesses compared to IBN. As shown in the table, the evolution of network automation paradigms has been significantly influenced by advancements across various fields. In particular, recent breakthroughs in Natural Language Processing (NLP) and Large Language Models (LLMs) should enable the use of human-readable structures and facilitate a transition to natural language for expressing intents [16].

In this work, we advocate the need for more experimental testbeds, particularly for developing cobot use cases. Nevertheless, in recent years, we have seen a variety of testbeds appear both from the research community and the

TABLE I
COMPARISON OF NETWORK AUTOMATION PARADIGMS VS. INTENT-BASED NETWORKING (IBN)

Paradigm	Description	Strengths vs. IBN	Weaknesses vs. IBN
CLI Scripting	Low-level command automation via scripts (e.g., SSH).	Simple, direct device access; familiar to operators.	Low abstraction, poor scalability, error-prone, lacks verification.
Model-Driven (YANG/NETCONF)	Structured configuration using data models and protocols.	Deterministic, validated configs; fine-grained control; vendor-neutral.	Requires vendor support; steep learning curve; lacks closed-loop.
Declarative (Ansible, etc.)	Defines desired state using templates/playbooks.	Easier than scripting; integrates with CI/CD; reusable artifacts.	No reasoning about intent; lacks autonomy and assurance.
Policy-Based	Rule-based control over specific behaviors (e.g., QoS, firewall).	High-level abstraction; aligns with SLAs; domain-specific.	Limited to predefined scopes; static; less flexible than IBN.
AI/ML-Based	Closed-loop automation using analytics and ML.	Proactive, adaptive; supports anomaly detection, self-healing.	Data-dependent; hard to explain decisions; complex deployment.
DevOps/GitOps	Infrastructure and service lifecycle managed as code.	Cloud-native, scalable, modular; aligns with CI/CD pipelines.	Focus on infra, not business intent; requires manual oversight.
Digital Twins	Simulated digital replicas for planning/testing.	Great for "what-if" scenarios; safe testing; enhances observability.	High modeling complexity; still emerging; needs real-time sync.

industry. For instance, Cisco has defined a road map for implementing IBN functionalities in its network deployments [17]. As another example from research efforts, the VITAL-5G project has deployed testbeds in transport and logistic sites (T&L) such as logistic centers, shipyards, and docks in multiple European countries [18]. The project aims to optimize operations by providing a service portal to bridge the gap between network management and business verticals through Intent-based APIs and platform-agnostic network applications (NetApps) [18].

In another recent work [19], the authors developed an IBN framework designed for the automated deployment of QoS-aware and SDN-enabled network slices in vertical service domains. The proposed framework, integrated with an NFV orchestration stack and the MANO platform, allows tenants to specify high-level connectivity and service requirements through intents. The system architecture includes key components such as the Intent Engine, Translation and Execution, Slice Management, and Monitoring. Experimental results in an emulated Metro SDN network demonstrate its feasibility and effectiveness in dynamically deploying and managing network slices.

Another case, where intents are used within the SDN context, is the work in [20], where the authors demonstrate the use of a novel IBN engine to interact with a Packet Optical SDN controller including Natural Language Processing to translate user commands to control and manage optical network operations. The solution presented offers an interactive User Interface (UI) that allows a user to interact with the system in an agile and smooth way without the need to know the optical resources in the infrastructure below.

In [16], Mekrache et al. introduce an Intent Life-Cycle (LC) management architecture designed to orchestrate and manage network services using large language models (LLM). They report experimental results of one of their components implemented in a 5G platform. However, they recognize that the development of their proposal faces several challenges. In a closely related work, [21], the authors propose a policy-based approach to decompose intents into a hierarchy of policies and a CCL automation module, based on FSM to implement the intents. They have evaluated their proposal through a cloud intent use case comprising a VNF and a health check service. The authors conclude that there are still many challenges including those related to the domain-specific execution, i.e., use cases.

From a cobot use case perspective, the VDA 5050 standard addresses key interoperability challenges in production environments. These challenges arise when automated guided vehicles (AGVs) from different manufacturers operate together. Traditionally, each manufacturer used its own master control system, necessitating additional coordination layers when integrating AGVs from multiple brands. VDA 5050 resolves this by introducing a standardized communication interface between AGVs and a central master control system, eliminating the need for proprietary control systems [22]. The standard allows engineers to accommodate AGVs with different autonomy levels and benefits decision-making capabilities for navigation (e.g., route selection, intersection handling). However, in our work, we extend these capabilities further by integrating intent-based service management, virtualization and orchestration capabilities into robot operations. Transitioning from a closed proprietary system to an open, cloud-based orchestration framework unlocks new possibilities for scalability, automation, and dynamic resource allocation.

This work extends our previous works [23], were the importance of Closed-Loop (CL) cobot control is emphasized in the context of 6G networks. A framework for implementing CL systems is proposed, with an experimental setup and a demonstration of a CL-enabled application migration use case. Offering significant insights into the intersection of 6G technologies and cobot networks.

III. USE CASE AND SYSTEM ARCHITECTURE

Imagine a scenario where an industrial facility manager wants to deploy a cobot surveillance service in a warehouse. The premises may house sensitive materials and machinery that require thorough supervision and examination, and static surveillance cameras are insufficient. Additionally, the manager wants the robots to use advanced image classification

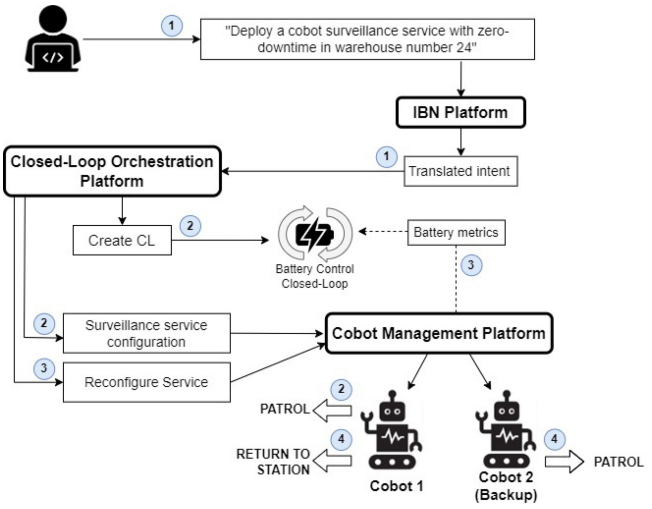


Fig. 1. High-level workflow of the cobot service management framework assisted by the closed-loop orchestration platform and the IBN platform.

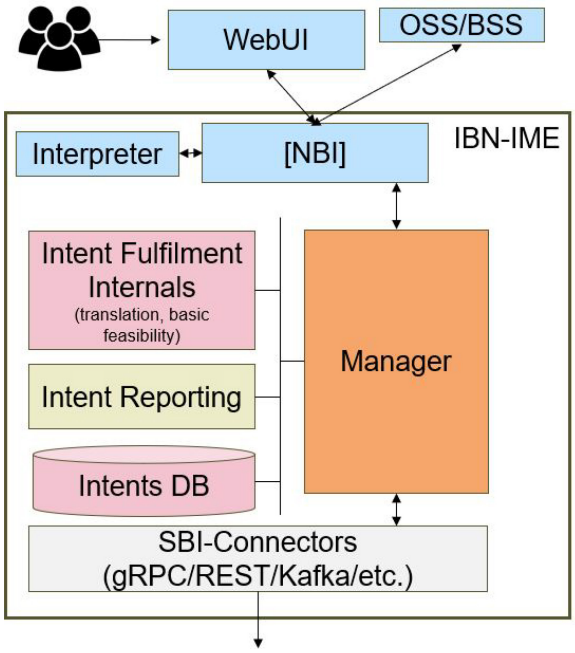


Fig. 2. IBN-IME internal architecture.

algorithms to assess the condition of the industrial assets, as small details might be missed by the human eye or not monitored by sensors.

Our proposal consists on the use of several collaborative robots (cobots), working in tandem to fulfill the business intent, that is, ensure that the business premises are under surveillance with no interruptions. This work develops and evaluates the implementation of the surveillance intent, where the ultimate goal is: 1) to analyze AI solutions enabling the implementation of the translation and decomposition of intents; and 2) define and validate the effectiveness of closed-control loops enabling the fine-grained control over the operation of the cobots.

A. Surveillance Cobot Use Case Scenario

This section focuses on presenting the design and operation principles of the cobot use case illustrated in Fig. 1. As seen from the figure, its main components consist of an IBN Platform, the Closed-Loop orchestration Platform and a Cobot Management Platform.

The use case is demonstrated by monitoring the life cycle of the intent instantiation and deployment and testing, the resilience of the system stack due to changes which may affect operations, see Fig. 1. The main events that describe the use case scenario are:

1) A business intent is submitted via the IBN platform in natural language. The platform interprets and resolves ("translates") the intent into a data model that specifies the required services and configurations to declare the desired system state.
2) The Closed-Loop (CL) Orchestration Platform receives this model, detailing the services and requirements. It initiates the surveillance service by activating cobot patrol and video streaming, and deploys a CL function to monitor cobot battery levels to ensure uninterrupted operation (zero-downtime).
3) The Cobot management platform reports the cobot battery levels to the CL. When a low the battery is detected, the Battery Control CL notifies the CL Orchestration Platform of the corrective action that needs to be taken. This action prompts a service reconfiguration.
4) The reconfiguration replaces the active cobot, with a fully charged backup. The first cobot returns to the charging station, while the second begins patrolling. Additionally, the video streaming will stop running on the first cobot and start on the replacement cobot to continue the surveillance mission with no interruptions.

This constitutes the main functionalities and objectives that the PoC implements. The following subsections detail the internal architecture of the modules that comprise the use case architecture.

B. IBN-IME Platform Architecture

To describe the principles of operation and modules of the IBN-IME platform, the selected intent example expressed in natural language is the following *"create a cobot application on warehouse 24 with zero downtime"*.

Fig. 2 shows the internal architecture of our IBN platform, denominated from now on as "Intent-Based Network-Intent Management Entity" (IBN-IME). Its design is based on the specifications of the "Intent Translation and Provisioning" enabler introduced in [24]. Its architecture is in constant evolution toward continuous improvement in various aspects, such as the integration of an NLP user interface and the incorporation of a larger number of domain resources and controllers.

The user enters the intent request via the WebUI. The WebUI forwards the request to the NBI, which must identify whether the incoming request has a human-based shape or a standard intent-based data model format, such as the one defined by the 3GPP [13]. If the request is already in standard format, the NBI forwards it directly to the Manager module

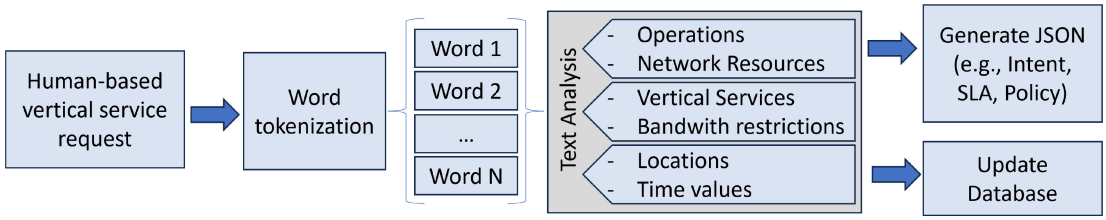


Fig. 3. NLP algorithm implemented.

for its processing. In the case of human-based requests, they are forwarded to the Interpreter module, which implements an NLP processor. Its primary task is to identify the keywords that define the expectations and targets constituting the user's intent.

Fig. 3 depicts the NLP algorithm implemented by the Interpreter and whose main result is to create a 3GPP intent data model. The NLP algorithm should be able to identify the different expectations composing the intent with its associated contexts as defined in [13]. As seen from the figure, the intent request is first tokenized into words. Then, a keyword search is implemented with one-hot encoding searching for relevant terms such as the operations, the locations, the vertical services, possible bandwidth restrictions and other attributes referring to different parts of the network. The extracted information out of the user request is used to construct the high-level intent. In the case of our sample intent, the following information is extracted from the human-language expressed intent:

- Identified operation: create.
- Identified resource: cobot application.
- Identified locations: warehouse 24.
- Identified resilience: True (zero downtime requested).

At this time, the Interpreter proceeds to create the first blueprint of the intent request based on the information provided by the user. All this information is forwarded to the Manager, which stores the intent data model shown in Fig. 4.

Once the blueprint created, the Intent Fulfilment module applies a feasibility check to verify that the expectations composing the intent are viable. This module consists of a set of functions required to achieve the complete management of intents. Besides the feasibility process to check if an intent may be accomplished, the Intent Fulfilment module should identify the executable actions to achieve the complete intent provisioning and the translation of those executable actions into the right and specific requests to publish in the Management Capabilities Exposure Framework (MCEF) for the right domain resources managers or controllers.

If the intent is not feasible, the user is notified and the intent is canceled. On the contrary, if the intent is feasible, the Manager identifies and maps the expectations into the specific deployment actions. The Manager instructs the different resource controllers through the SBI, generating the multiple and necessary requests for each involved resource controller. The SBI then establishes communication with the service orchestration module to initiate the deployment of the requested service, including the corresponding CL.

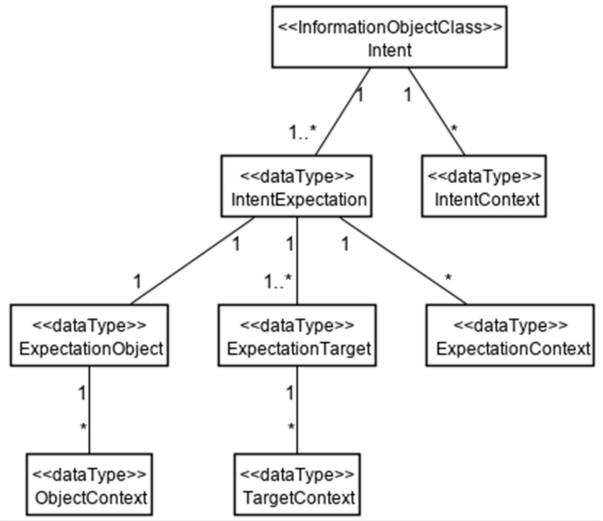


Fig. 4. Intent data model.

It is worth noting that the role of the Intent reporting module isn't limited to generating the intent's status report, but it also stores the intent data model into the database.

C. Service and Closed-Loop Provisioning

The orchestration of the service and its dedicated CL are triggered by the Closed-Loop Orchestration Platform, see Fig. 1. In this section this component is expanded to show its internal architecture. Fig. 5 shows the interactions among the main platform components when deploying the service declared by the intent and its corresponding CL. After receiving the Intent Data Model, the Service Orchestrator instructs the Resource Orchestrator to provision a video-streaming surveillance application on an appropriate cobot. From the platforms perspective, the Resource Orchestrator is in charge of directly handling all the platform's interactions with the cobots. Therefore once the request is received the Resource Orchestrator starts a three-step execution process: (i) selecting a suitable cobot, (ii) deploying the video-streaming application on the selected cobot, and (iii) requesting the patrolling task by sending a specific command to the target cobot via a message broker.

In parallel, while the service orchestration is happening, the Service Orchestrator also requests the deployment of a proper CL for the target service. This is done through an interface exposed by the CL Governance, specific for the CL life-cycle management. As illustrated in Fig. 5, upon receiving the request from the Service Orchestrator, the CL Governance

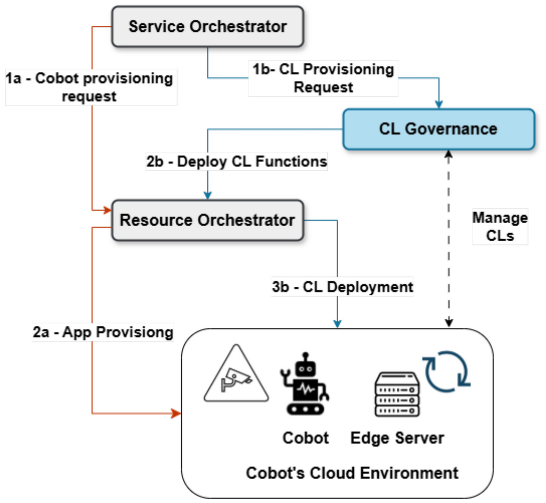


Fig. 5. Service and CL deployment with involved entities interactions.

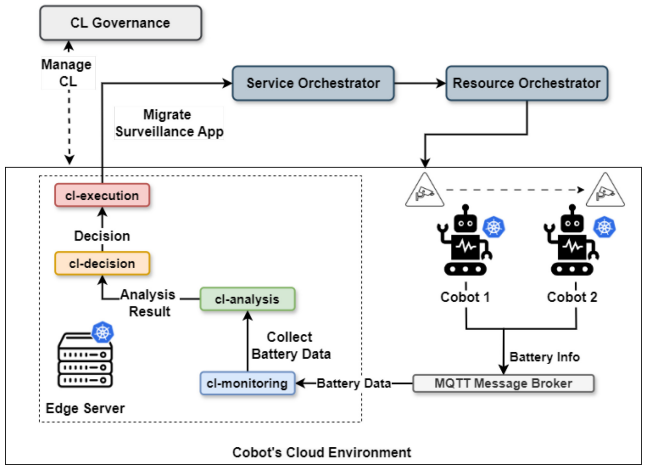


Fig. 6. CL setup on Cobot's cloud environment.

selects the proper CL descriptor from its internal catalog and identifies the target cloud platform for the deployment. At this point, the CL Governance requests the deployment of the CL Functions.

The CL is reactive and tuned to take decisions based on the battery level of the cobots: That is to say, if the battery level of the cobot drops below a given threshold, the CL reacts tuning the service in order to select a new suitable cobot. Fig. 6 depicts the CL Functions in action. A Monitoring stage (cl-monitoring) collects the cobots battery level to then forward this information to an Analysis stage (cl-analysis). If the battery level is deemed insufficient to continue the surveillance task, the CL requests the Service Orchestrator to update the service during the Decision and Execution stages (cl-decision and cl-execution). In this use case, the update consists on dispatching a new cobot and migrating the video surveillance application from the current cobot to the new one.

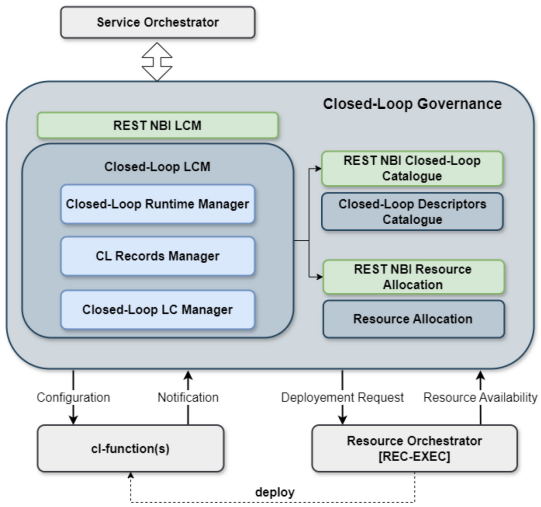


Fig. 7. CL Governance high-level architecture and interactions.

D. Closed-Loop Automation and Coordination

Our proposal represents a modern conception on the implementation of CLs, where their different stages are deployed and configured on demand. The management happens through the CL Governance, one of the two CL management functions defined by ETSI ZSM in [11]. The CL Coordination, the second management function, aims at coordinating multiple CLs relying on the same set of resources. This function is not demonstrated in this work and the section mainly focuses on the CL Governance, originally detailed in [25].

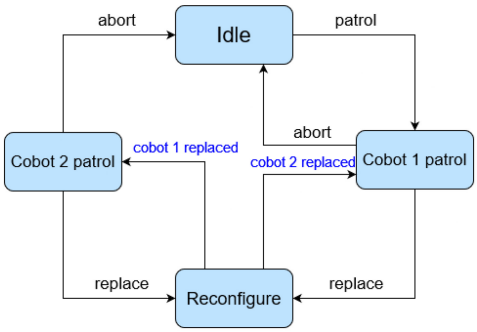
Fig. 7 shows the main architectural components of CL Governance module, along with the interactions with the other entities required for the service orchestration.

The CL Governance implements several functionalities for the management of the CL. In particular, the life-cycle management is realized through three different modules:

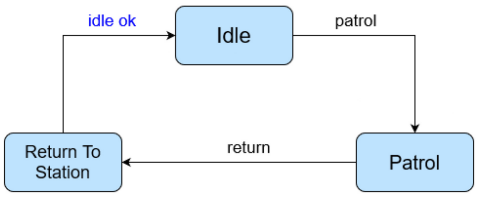
- Closed-loop Life Cycle Manager. In charge of selecting specific CL and CL functions' descriptors from the catalog (see below) given the service requirements. It also provisions and decommissions CLs. The Service Orchestrator exploits its interface to request the provisioning of CL instances.
- CL Records Manager. Maintains the runtime information of provisioned CLs.
- CL Runtime Manager. Provides the logic to perform runtime operations on a given CL e.g., start/stop/pause CL, configuration, etc. It offers a dedicated REST-based NBI for CL coordination.

The functionalities implemented for the CL life-cycle management are supported by two additional modules:

- Closed-Loop Descriptors Catalogue. Manages the descriptors for both the CL and the related functions. The CL is indeed realized as a cloud-native application formalized through specific information models.
- Resource Allocation module. Exploits information from the Resource Orchestrator to compute the placement of the different CL functions: it may happen on different places (cloud, edge-cloud, far-edge platforms) depending on the CL requirements.



(a) Cobot Service Manager State Machine



(b) Cobot State Machine, for an individual cobot

Fig. 8. State Machines of application components of the cobots and the cobot service manager.

E. Cobot Management Platform

Another key aspect of this testbed is the definition of state machines (SM) needed for a safe cobot coordination in the premises, all while ensuring zero service downtime. We have followed a hierarchical state machine approach allowing a fine-grained control over the cobot tasks. The PoC includes three state machines, two for the each one of the cobots (Cobot 1 SM and Cobot 2 SM) and one master SM (Cobot Service Manager) that coordinates the subordinate state machines aforementioned. Each SM is defined statically, that is, the states, movement and actions that the cobots perform have to be specified by an engineer so the cobot can operate safely in the premises.

Fig. 8, shows the state machines definition and the relation between the Cobot Service Manager and an individual Cobot State Machine. While in the *Idle* state, the Cobot Service Manager is ready to accept service requests from the CL Platform. Upon the arrival of a service request, the service moves from the *Idle* state to the *Cobot 1 patrol* state. In case that the cobot is unable of providing the service other than for the energy constraint, the service provisioning is aborted. Otherwise, the cobot operation initiates its operation as specified by its SM. Fig. 8b, represents the SM of the cobots. In short, the Cobot Service Manager SM states *Cobot 1 patrol* or *Cobot 2 patrol* trigger the transitions in the respective Cobot SM from the *Idle* state to the *Patrol* state. While in the *Idle* state, the cobot is ready waiting in the charging station. The *Patrol* state indicates that the cobot is navigating the patrol waypoints in a loop. However, while in the *Patrol*, the cobot must monitor the energy level of its battery. As seen from Fig. 8a, the *Reconfigure* state coordinates two state transitions, one cobot returning to the station (*Return To Station* state) and another starting to patrol. In our use case, the actions associated to this transition are challenging due to the limited space available to perform this maneuver. That is to say, the navigation plan of the cobots should make sure that they do not collide with each other. Therefore, once a replacement is ordered, the Cobot Service Manager inspects the last notified location of the cobot to be replaced, and provisions the replacement accordingly to minimize downtime while avoiding the risk of collision.

IV. Experimental Setup and Testbed

The hardware and software that support the use case are as equally important as the enablers presented in the previous section. Fig. 9 shows a detailed schematic representation of the PoC. Careful thought has been put into the setup of the edge servers, cobots, network and the integration of the enabler modules (IBN-IME and CL Orchestration Platform). In the upper half of the figure, the edge servers and the Workstation contain the modules in charge of: orchestrating services (Kubernetes master node and CL Platform), operate the cobots (State Machines) and monitor the testbed (MQTT broker and a RTMP video server). In the lower half, the cobots software stack is presented along with various sensors, modems and other hardware components attached. The Ethernet LAN and the 5G SA network, implemented with a Open5Gs core and a Nokia ASiR RAN, provide connectivity between all the modules.

The icons inside the big boxes give an insight into the software architecture. Blue heptagons represent Kubernetes resources, under the dashed lines the Kubernetes control plane and over the lines the resources deployed can be observed. Note that cobots are implemented as far-edge nodes (worker or agent node) and, from the orchestration perspective, this enables the video surveillance service migration use case.

The Workstation comprises the cobot state machines, the IBN-IME and the CL Platform. All the software development related to robotic operations have been implemented using the Robot Operating System (ROS) framework. The state machines, implemented as Docker containers, connect to the ROS Middleware module (also known as ROS Master) located in the cobots. This module operates the motor, wheels and the LiDAR sensor shown in Fig. 10, and enables autonomous navigation in the premises. Therefore, the state machines simply indicate the waypoints the cobots must follow while operating in *Patrol* and *Return To Station* states as shown in Fig 11.

The IBN-IME and the CL Platform are implemented in dedicated virtual machines. The use case user interacts with the IBN-IME through a Web UI, and the later sends the data model to the CL Platform through Rest APIs. The CL Platform can orchestrate services in Kubernetes (eg. RTMP Video Client) and deploy dedicated CL instances depending on the intent, as shown in the upper-left orange box in Fig. 9.

Moreover, the MQTT broker, deployed in the Edge Servers, has a dual purpose. Firstly, the CL monitoring function subscribes to the broker for collecting battery metrics from the Battery Reader modules running in the cobots. Secondly, the CL Resource Manager can publish MQTT messages that

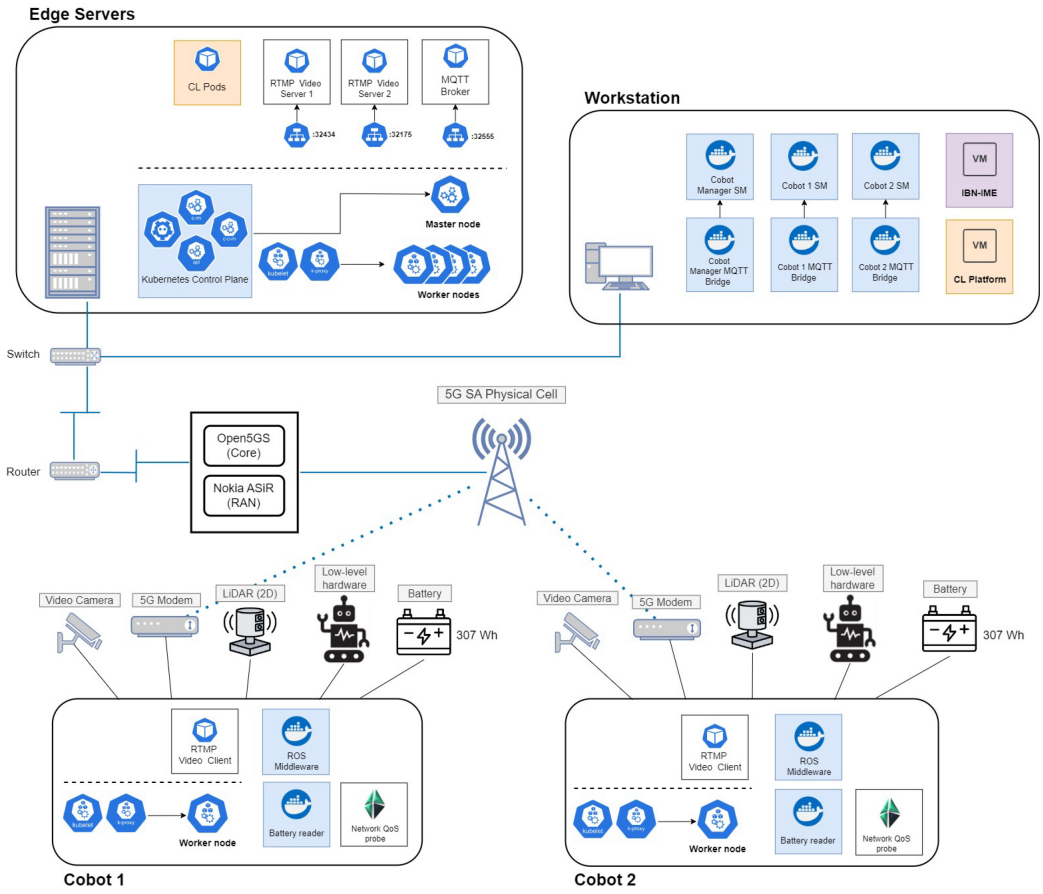


Fig. 9. Implementation overview of the cobot service management use case with the deployed hardware, software and networking components.



Fig. 10. Physical cobot setup, showing one of the two task workers used in the PoC.

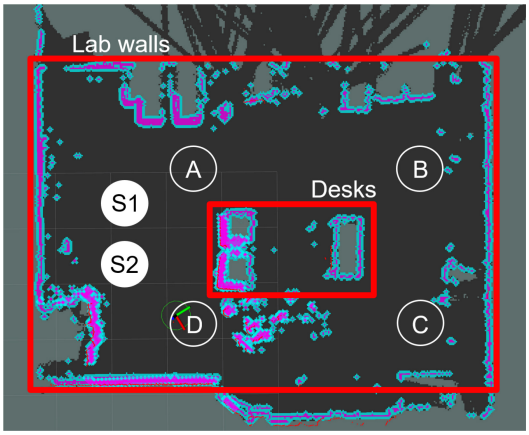


Fig. 11. ROS map for cobot navigation. Surveillance waypoints (A, B, C, D) and charging stations (S1 and S2) are marked, as well as the limits of the lab premises.

trigger state transitions in the SM presented in Section III-E. This is possible thanks to the ROS to MQTT bridge modules in the Workstation, that translate MQTT topics to ROS topics and vice versa. Therefore, the MQTT broker acts as a central point of communications between the all the applications.

V. Results

This section presents the performance evaluation of the 6G enablers described in previous sections. Relevant metrics include systems intent responsiveness and instantiation times, and the video migration performance. These results are highlighted to validate the feasibility and effectiveness of the enablers in real-world use cases.

A. Intent-Based Service Provisioning Results

To evaluate the performance of our proposal, 15 trials were conducted. In each trial, the intent's requests were entered using human-like sentences to simulate a comprehensive

```
{
"request": "create a cobot service on warehouse 24
    with zerodowntime",
"observation_period": "0",
"intent_admin_state": "ACTIVATED",
"user_label": "cobot",
"intent-id": "ebb3c42e-ef5b-4588-a550-7bbe3c4412f7",
"name": "cobot_warehouse_24",
"intent_priority": "50",
"intent_report_ref": "c10d602f-27a3-41a1-80ee-32
    ee7dd9fff3",
"intent_expectations": [
  {
  "expectation_object": {
    "expectation_object_type": "Network Service",
    "expectation_object_instance": "9a294312-ed73
        -4765-b473-178494c08c34",
    "object_contexts: [
      {
      "context_value_range": "['24']",
      "context_condition": "IS_EQUAL_TO",
      "context_attribute": "warehouse"
      },
      {
      "context_attribute": "soccer_service",
      "context_condition": "IS_EQUAL_TO",
      "context_value_range": "a17b28e4-4fb5-42df-
          b285-88b66e0c7a37"
      }
  ]},
  "expectation_id": "bcdad622-73f8-4787-af09-
      a0c31ddf203b",
  "expectation_verb": "DELIVER",
  "expectation_targets": {
    "target_contexts": []
  },
  "expectation_contexts": [
    {
    "context_value_range": "563fe86e-00ae-11ef-92c8
        -0242ac120002",
    "context_attribute": "sla",
    "context_condition": "IS_EQUAL_TO"
    }
  ]
  }
],
"intent_contexts": []
}
```


Listing 1. Intent Data Object Example


```
{
"intent_report_id": "c10d602f-27a3-41a1-80ee-32
    ee7dd9fff3",
"intent_ref": "ebb3c42e-ef5b-4588-a550-7bbe3c4412f7"
    ,
"last_updated": "2024-05-30 10:20:13.598",
"intent_fulfilment_report": [
  {
  "intent_fulfilment_info": {
    "not_fulfilled_state": "",
    "fulfilment_status": "FULFILLED",
    "not_fulfilled_reason": ""
  },
  "expectation_fulfilment_results": [
    {
    "expectation_id": "bcdad622-73f8-4787-af09-
        a0c31ddf203b",
    "expectation_fulfilment_info": {
      "not_fulfilled_state": "",
      "not_fulfilled_reason": "",
      "fulfilment_status": "FULFILLED"
    },
    "target_fulfilment_results": []
    }
  ]
  }
],
"intent_conflict_reports": [ {} ],
"intent_fasibilitycheck_report": {
  "infeasibility_reasons": "",
  "feasibility_check_result": "FEASIBLE"
}
}
```


Listing 2. Intent Report Example

deployment scenario. As a main result, the corresponding provisioning times associated with the intent-based deployment process were obtained.

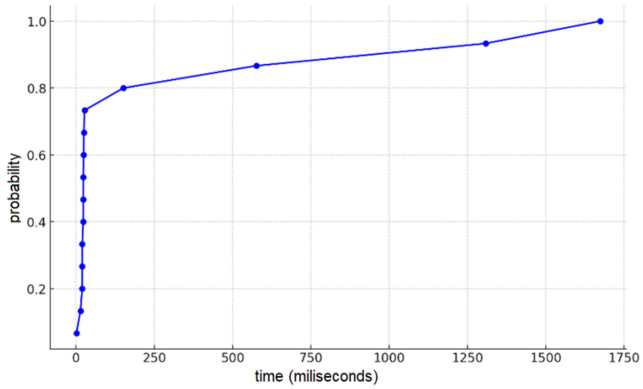
Listing 1 shows the generated intent-based data object based on the human sentence: "create a cobot service on warehouse 24 with zerodowntime". As shown in Listing 1, once the NLP is applied, the intent object is generated with a list of expectations and targets. In this case, there is just one expectation of type "Network Service". The expectation is to be delivered by the Service Orchestrator in Continuum of Cloud and Edge Resources (SOCCER, introduced simply as the CL Service Orchestrator in Section III-C). The intent object also specifies the expectation's location and the reference to the SLA selected according to the quality of service requirements (i.e., zerodowntime/resilience).

In addition to the intent data object, an associated intent report is also created following the 3GPP data model specifications [13]. As shown in Listing 2, the first parameter, "intent_ref", corresponds to the identifier of the associated intent. The report consists of three main blocks: a) "intent_fulfilment_report" that includes the individual status of each expectation (and their targets) composing the intent, b) "intent_conflict_reports" refers to the list of possible conflicts that may appear during the life cycle of the intent, and c) "intent_feasibilitycheck_report" which states whether the generated intent is feasible. If the intent is deemed infeasible, this section also includes the reasons for its infeasibility.

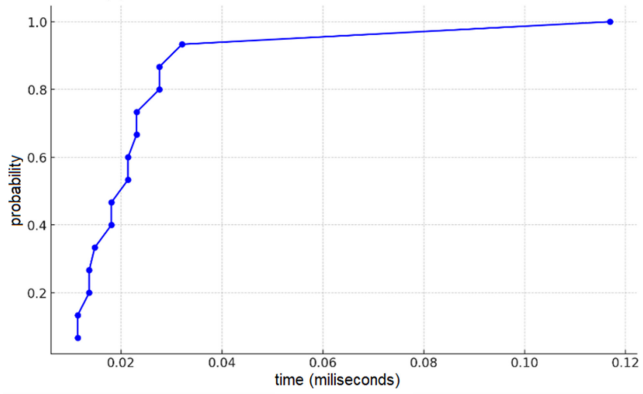
In each trial, we deployed various probes to measure the total time required for the deployment of the surveillance cobot application and the execution times of two of its main tasks, namely the NLP algorithm and the intent feasibility process. One of our main goals was to verify the impact of the NLP algorithm and feasibility test over the total time to deployment.

Fig. 12a shows the cumulative distribution function (CDF) of the time to deploy the service. As seen from the figure, 11 out of the 15 deployment trials were performed in less than 30ms. The other four deployment trials reported values in the range from 151.35ms to 1673.52ms. In other words, we observe that the mean service instantiation time of our testbed is 363.75ms. These results show that the implementation of our proposal can effectively provide the deployment of the cobot surveillance application in less than 2s.

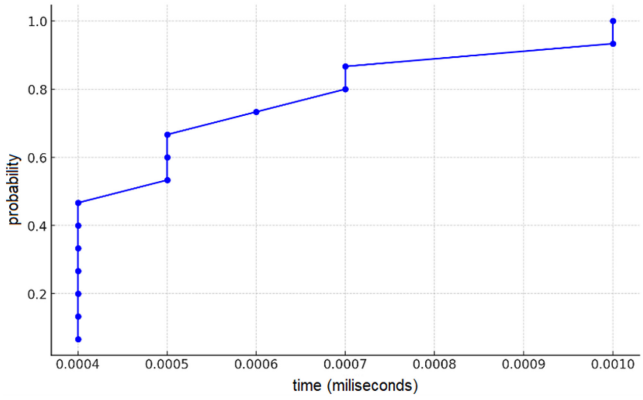
Figs. 12b and 12c show the NLP and intent feasibility processing times. For both metrics, all 15 samples show very low values, with a maximum of 0.1ms for the NLP algorithm and 0.001ms for the feasibility process. These results clearly indicate that the main factor delaying the instantiation of an intent is the actual deployment and configuration of the



(a) Vertical service instance creation task CDF.



(b) NLP task CDF.



(c) Intent feasibility task CDF.

Fig. 12. CDF for IBN-IME tasks.

services over the physical resources. Although the NLP and feasibility verification processes don't significantly contribute to the overall latency in the current scenario, this may not be the case in more complex scenarios. In particular, if different domains and/or providers with similar resources coexist, the IBN-IME may need to apply a longer feasibility process. This process would be followed by a more complex resource selection.

B. CL Assisted Zero-Downtime Service Migration Results

Once the surveillance service is deployed—meaning a suitable cobot is selected, provisioned with a video streaming application, and the dedicated closed loop is operational—the Monitoring stage continuously monitors the cobot's battery

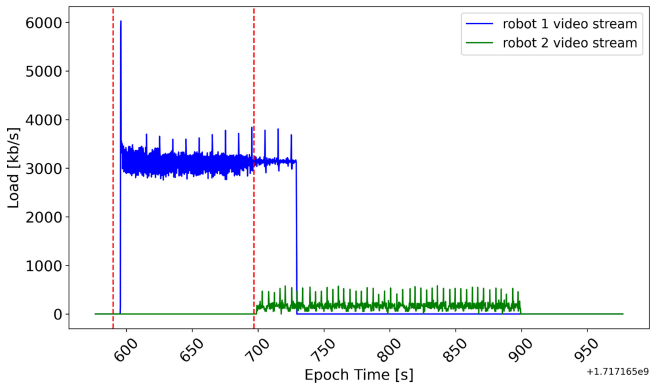


Fig. 13. Video service throughput during deployment and migration (from cobot 1 to cobot 2) with their respective timestamps marked in red.

status. As mentioned, the closed-loop here considered is reactive, i.e., it reacts to a certain event, which is in our case the battery level goes below the established threshold. When the positive event is detected, the reaction time of the CL is in the order of milliseconds. The Analysis stage retrieves battery level samples each second, buffers them and calculates the average over 20 samples. This logic purposefully ignores potential false positives, preventing unnecessary CL executions and consequent service reconfiguration. Indeed, one thing to take into account is that isolated spikes and fluctuations are possible and induce the risk of false positives, especially when the battery level is approaching the given threshold, in this experiment fixed at 15%. The Decision stage checks the average values received by the Analysis with the threshold and when it is crossed, a service migration towards another cobot is decided and timely requested to the Service Orchestrator through the Execution stage. Note that after the migration decision is taken, the Decision stage starts receiving higher battery values. The service reconfiguration also includes a CL update since now the battery to be monitored is from the replacement cobot.

The reactive CL operation can be observed from the traffic generated by the video surveillance application during all the stages. Back in Fig. 9 a key component to the use case evaluation was left without a proper introduction. The Network QoS Probes, running Qosium [26] software, allow us to accurately measure simultaneously the traffic flows between the cobots (RTMP video source) and the Edge Cluster Servers hosting the RTMP video servers. Since our proposal should deliver a zero-downtime video surveillance service, we focus now on examining the operation of the video transmission from both cobots. In other words, it is important to verify that in fact the video surveillance service remains active from the moment it is launched, during the service migration from one cobot to the other one and up to the end of the surveillance patrol. Any interruption could create a vulnerability that might be exploited. Towards this end, it is essential to verify the operation of the CL processes including the video service implemented on each cobot.

Fig. 13 shows the video service throughput over the life span of a surveillance mission. The red vertical lines mark the time instants when the CL Platform issues the commands.

In the figure, the first red vertical line shows the time instant when the first patrol command is issued by CL platform. Three seconds later the surveillance application starts transmitting the video stream from Cobot 1 to the server. The second red line indicates the time instant when the migration from Cobot 1 (blue) to Cobot 2 (green) is initiated. This process is activated upon receiving 20 consecutive messages of low battery metrics from Cobot 1. The CL dispatches the command to replace Cobot 1 by Cobot 2. That is to say, the CL instructs Cobot 1 to return to the charging station and transfers the video surveillance service to Cobot 2. Less than one second afterwards the video streaming application is already up and running on Cobot 2. Nevertheless, we can observe that the video stream from Cobot 1 takes a little longer to stop despite having been instructed to do so. As seen from the figure, the video streaming from Cobot 1 only finishes the transmission after 25 seconds. This is due to the video data found in the internal buffer of the video system deployed on Cobot 1. Finally, the patrol mission finishes when CL sends a termination instructing Cobot 2 to return to the station, terminating the video stream.

Note that each cobot is using a different video device. The Logitech webcam used by Cobot 1 support h264 encoding while the Intel RealSense uses yuyv422. This explains the difference in network load produced between both devices.

VI. Discussion and Future Work

The following section highlights issues that, while not central to the use case presented in this work, are essential for developing a complete technological solution. It also points out to some of the challenges currently being explored by HEXA-II.

A. Intent Conflicts Management and Negotiation

An essential part of the intent life cycle is conflict management, which involves detecting and resolving issues during intent requests and provisioning. Effective conflict management depends on identifying the types of conflicts and applying suitable detection and resolution methods. Conflicts can occur either within a single intent, such as conflicting expectations or targets, or between multiple intents when their goals or required resources clash.

A common approach for conflict detection is constraint-based algorithm selection, which uses specific parameters to identify potential issues. For instance, a conflict might arise if several gaming services are requested simultaneously, but system resources are insufficient to support them all. For resolution, strategies like priority-based policies or optimization-driven methods can be applied, depending on the context.

In environments with multiple stakeholders, conflict likelihood increases due to the wide variety of requests coming from different users and industries, especially at the business level. Handling these conflicts effectively requires careful system design and robust negotiation mechanisms. When a conflict is detected, the system must either resolve it by adapting the incoming or already deployed intent, or notify the requester [27].

Another critical feature of a fully functional intent-based networking (IBN) system is the ability to negotiate the reformulation of intents. This involves assessing available resources, similar to what is done during feasibility checks, and generating alternative requests the user can choose from. The main challenge in this negotiation process is not identifying available resources, which is already handled, but rather interacting with the user—particularly when the original intent was expressed in natural language. To address this, some solutions incorporate chatbot interfaces to support intent reformulation and conflict resolution through more natural, user-friendly communication [28].

B. Closed-Loop Orchestration

The closed-loop (CL) presented in this work is implemented as a cloud-native microservice application, with each microservice representing a distinct stage of the loop. This architectural approach enables the integration of automation mechanisms tailored specifically to the target environment and the intended automation objectives. Through this setup, the CL Governance function is capable of selecting the most appropriate composition of stages.

In the demonstration scenario, the closed-loop has been customized to support a video surveillance service, operating solely at the service layer. Here, any corrective actions are implemented by modifying the service itself. That is to say, it is triggered as a result of service provisioning initiated by the IBN-IME through the service orchestrator.

The concept of dynamic closed loops—capable of being orchestrated, reconfigured at runtime, and tuned for specific tasks—is envisioned as a critical component of the pervasive automation expected in 6G networks. This model raises several challenges. Among them is the potential conflict arising from the coexistence of multiple closed loops operating within the same environment. Such overlaps may lead to race conditions or contradictory actions, resulting in system inconsistencies or instability that could undermine the very benefits of automation.

The ETSI ZSM framework has examined these concerns and, as outlined in [11], introduced the concept of CL Coordination. This mechanism is intended to prevent or mitigate conflicts by orchestrating decision-making across closed loops. It also supports various forms of collaboration between loops to enhance the overall efficacy of the automation framework. These collaborations may follow either a peer-to-peer or hierarchical model. In hierarchical systems, two modes of interaction are typically considered: escalation, where a lower-level loop requests a higher-level loop to resolve a shared objective; and delegation, where a higher-level loop entrusts a lower-level one with making automation decisions.

Further developments, such as those proposed in [21], explore the use of hierarchical closed loops within intent management systems. These systems leverage delegation to ensure the fulfillment of user intents and could be instrumental

in enabling the dynamic, on-demand configuration of cloud-native closed loops.

C. LLM User Interfaces and AI CL

The IBN-IME solution described in this work—is an evolving prototype still under active development within the Hexa-X-II project [5], with further enhancements planned in upcoming research initiatives. One of the key design decisions in its current iteration was to employ Natural Language Processing (NLP) rather than more complex AI/ML technologies such as Large Language Models (LLMs). This choice was driven primarily by the simplicity and agility offered by NLP. Unlike LLMs, which require extensive input formatting—composing a message with all relevant requirements and relying on a set of predefined examples and attributes—NLP operates through a straightforward synonym-based interpretation of incoming sentences. This allows for faster, more lightweight processing during early development stages.

Nonetheless, there are research efforts towards integrating LLMs into future iterations of IBN-IME, as outlined in [29]. This research reflects a broader ambition to expand the capabilities of the IBN-IME system.

Currently, the IBN-IME implementation employed in this work focuses primarily on intent interpretation and feasibility analysis. It does not yet address more complex challenges such as conflict detection, intent negotiation, or resolution mechanisms. Future development goals include enhancing the system to handle these additional tasks—specifically, identifying and resolving conflicts, enabling negotiation when an intent is deemed infeasible or leads to contention, and leveraging advanced AI/ML techniques to support these functionalities more effectively.

Similarly, the CL-based automation mechanism will be evolved in other similar activities. Future works may involve the interaction with specific MLOps platforms to request and deploy AI/ML-driven Analysis/Decision stages, evolving from reactive to predictive CLs. Additionally, as mentioned, the adoption of a mechanism capable of decomposing Intents into a set of specific CLs may enable Intent automation on the IBN-IME platform presented here.

D. Intent Based Frameworks in Cobot Operations

In a real warehouse environment, large equipment frequently moves in and out, altering the areas where cobots can navigate and work. As a result, cobot use cases, and the IBN frameworks that support them, must be flexible enough to adapt to physical changes in everyday operations. For instance, the reference map shown in Fig. 11 may not always resemble the scanned layout that the cobot senses in real-time. As mentioned in Section II, the VDA 5050 standard promotes standard interfaces and communications in robotics. Adopting a standardized message format would help accelerate the integration of intent-based management frameworks into cobot applications.

In future work, the VDA 5050 standard could be considered in the testbeds evolution. Its inclusion can improve the CL Platform's situational awareness. For instance, the platform could request updated maps when the physical layout changes, or assign tasks to cobots based on their individual capabilities as specified in the standard [22].

Additionally, Joint Communication and Sensing (JCAS) technologies are emerging as a key component in 6G development. These technologies could offer redundancy and real-time updates about changes in the environment, improving the reliability and adaptability of cobot operations.

While extending the given PoC, it is also required to take into account the security considerations of the given scenario. On one hand, the given scenario has the potential to extend for provisioning security services with closed-loop automation provided by the network and also for safeguarding 6G systems. On the other hand, it is also important to identify the possible security threats associated with the IBN enablers. Here ensuring intent integrity protection, access control, and assuring user privacy are some key properties to consider.

VII. Conclusion

The development of 6G networks represents a pivotal moment in the evolution of communication systems, as societal, environmental, and economic needs, coupled with current technological advancements, drive wireless networks beyond their original purpose. Inspired by the vision of the Hexa-X-II project, this work aims to demonstrate the workflow and performance of 6G enabler technologies through the implementation of a PoC. Cobot use cases are expected to play a significant role in bringing new value propositions to cellular networks.

The tests conducted in this PoC showcase the integration of 6G enablers, specifically the implementation of Intent-Based service provisioning over a 5G Standalone (SA) network. The workflows and interactions demonstrated in this paper are intended to serve as a precursor to further developments in 6G platforms and testbeds. Additionally, the measurements presented here highlight the feasibility of this system in delivering tangible value to industrial stakeholders. Only identifying the limitations in the PoC and through validation experiments can a full working solution be archived. Finally, having a good balance between full service automation and human management is a priority when safely orchestrating cobots in a physical environment.

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